400 COFFEE

Requirements Traceability & Testing

VERSION 1.1 | 06 NOVEMBER 2026

400 COFFEE

Requirements Traceability Matrix & Test Strategy

Functional coverage, performance targets, test cases and release evidence

Document status: Professional working draft for stakeholder review. Open decisions and assumptions are explicitly identified.

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1. Traceability approach

Requirement linkage Every functional requirement maps to one or more test cases, test level, owner, evidence and release gate. The accompanying XLSX contains an editable starter RTM.

Status values Not Started, In Progress, Pass, Fail, Blocked, Deferred. Deferrals require product owner approval and a documented release boundary.

Evidence Store build/version, environment, test data reference, timestamp, tester, result, defect ID and supporting logs/screenshots.

2. Functional test scenarios

TC-001 Tenant isolation Tenant A cannot read, update, export, subscribe to events for or infer Tenant B records.

TC-002 Catalog pricing Server total matches configured price, modifier, tax and charge rules; historical order snapshot remains unchanged after catalog edit.

TC-003 POS idempotency Retry same create-order request with same idempotency key; only one order and one set of station tickets exist.

TC-004 Dine-in/takeaway Both order types preserve required table/pickup metadata and correct downstream display.

TC-005 Station routing Mixed beverage/food order creates correct station tickets with no missing or duplicate items.

TC-006 Parent readiness Parent order cannot be ready until all required station tickets/items are ready.

TC-007 QR policy Cashier-held and auto-release policies behave as configured and policy snapshot remains on order.

TC-008 QR token security Invalid, expired or altered token cannot access unrelated order or staff data.

TC-009 UPI declaration Customer declaration does not mark payment paid or increase confirmed collections.

TC-010 UPI verification Authorized staff verifies merchant-side receipt; ledger records amount/reference/actor/time; payment state updates correctly.

TC-011 Duplicate payment reference Duplicate reference is rejected or sent to reconciliation exception; no duplicate collection is counted.

TC-012 Partial/overpayment Partial payment leaves balance outstanding; overpayment is flagged and does not silently distort totals.

TC-013 Refund Refund is recorded as separate ledger movement and net collections update correctly.

TC-014 Shift close Expected totals, counted values, variance and reason are stored and auditable.

TC-015 KDS reconnect Disconnect/reconnect recovers authoritative queue without duplicate fulfillment.

TC-016 Order cancellation Permitted cancellation propagates to all affected station tickets and preserves audit trail.

TC-017 Dashboard reconciliation Gross order value, verified collections, refunds, outstanding and net collections reconcile to source records.

TC-018 Role permissions Cashier, kitchen, manager, owner and platform operator actions match role matrix.

TC-019 Device sync Unsubmitted local draft survives refresh/restart where supported; server-acknowledged orders remain recoverable.

TC-020 Load/peak simulation Run at 3× measured peak with concurrent cashier/KDS/owner clients; capture latency, error rate and saturation.

3. Performance and reliability targets

API reads Proposed p95 <500 ms for ordinary reads under agreed load profile.

Order submit Proposed p95 <1 s excluding third-party payment provider latency.

Kitchen propagation Typical update <2 s from server commit to connected station display.

Availability Proposed 99.5% monthly service target, subject to hosting design, maintenance policy and contract.

Recovery Proposed RPO 15 minutes / RTO 4 hours, to be validated through restore drills.

Load profile Use measured peak orders/minute and concurrent devices; test 3× peak. Do not substitute daily volume alone for concurrency.

4. Test environments and data

Environments Local, integration, staging/UAT and production smoke. No production personal/payment data in non-production unless masked and approved.

Data sets Representative menu with modifiers, mixed station routing, multiple order types, payment states, refunds, shifts, tenants and time-zone boundaries.

Test controls Resettable seed data; deterministic clocks where practical; unique idempotency keys; isolated tenant fixtures; masked logs.

5. Defect severity and release decision

Critical Data leakage, lost/duplicated orders, false paid state, incorrect financial ledger, unauthorized refund or inability to recover service. Blocks release.

High Major workflow unavailable or material metric misstatement with no safe workaround. Blocks release unless formally risk-accepted.

Medium/low Limited impact with workaround; triage and document before release.

Go/no-go All critical tests pass; no open critical/high defects; reconciliation evidence approved; security/tenant tests pass; operational pilot sign-off recorded.

VERSION 1.1 — CUSTOMER EXPERIENCE ADDENDUM

Version 1.1 addendum — 06 November 2026. This addendum supersedes any earlier customer-facing UX description that implied a conventional QR menu/dashboard as the primary customer experience. The immersive Virtual Café is the approved customer-side direction. Existing staff POS, KDS, owner and platform-admin interfaces remain separate workstreams and are not redesigned by this customer-side change.

6.1 Customer journey test coverage

CX-T01 first visit: entrance scene, manager greeting, menu and bill paths visible. CX-T02 reduced motion: camera and page transitions suppressed; all functionality remains. CX-T03 WebGL failure: 2D fallback renders; ordering succeeds. CX-T04 diary navigation: categories and back/forward preserve cart. CX-T05 product variants: selected size/modifiers and price deltas correct. CX-T06 instructions: notes persist into order and station tickets. CX-T07 unavailable item: cannot add or submit; clear explanation. CX-T08 server quote: manipulated browser price rejected/recalculated. CX-T09 phone validation and consent: invalid numbers blocked; consent audit captured. CX-T10 review edit: back navigation retains details. CX-T11 duplicate submit: repeated

click/retry yields one order. CX-T12 pay at counter: order created unpaid/due; cashier workflow receives it. CX-T13 UPI/QR: payment stays pending until server verifies. CX-T14 card: hosted provider flow; no PAN/CVV stored by café app. CX-T15 webhook: signature validation, duplicate event idempotency and amount mismatch. CX-T16 payment failure/expiry: clear retry/recovery. CX-T17 WhatsApp provider failure: order remains valid; notification failure visible/retryable. CX-T18 restaurant sync: order appears once; reconnect recovers missed event. CX-T19 bill lookup: wrong reference/phone does not disclose bill details. CX-T20 bill amount: client cannot alter server outstanding balance. CX-T21 mobile layout: 360px width, no horizontal scroll. CX-T22 accessibility: keyboard, labels, focus, contrast, screen reader. CX-T23 load/performance: scene assets lazy-loaded; checkout usable before high-quality 3D completes. CX-T24 analytics: no raw phone/PII in event payloads.

6.2 Visual quality gates

Compare against approved concept screenshots and design tokens at 360×800, 390×844, 768×1024 and 1440×900. Check typography hierarchy, café materials, lighting, manager composition, diary affordance, product imagery, price visibility, touch target size, contrast, safe areas, modal behavior and no layout shift. Visual regression should not freeze content prices or menu data. Any design change from the approved direction requires explicit product owner sign-off.

6.3 Payment and order invariants

A submitted order is not necessarily accepted. An accepted order is not necessarily paid. A payment attempt is not proof of payment. Payment can be confirmed only from server-verified provider status/webhook or authorized merchant-side manual verification. Amount/currency/order reference must match. Duplicate events and duplicate references must be idempotent or flagged. Refunds are ledger events, not deletion of original payments.

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